

Zhening ZHU

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EDUCATION

University of Leeds , Leeds, UK	2025.09 - Present
PhD Computer Science (EPSRC Doctoral Landscape Award Studentship) Topic: Perceptual graphics / bandwidth reduction using perceptual rendering	
Cornell Tech (Cornell University) , New York, USA	2024.08 - 2025.05
MEng Computer Science GPA: 4.0	
University of Nottingham , Ningbo, CN; Nottingham, UK	2020.09 - 2024.07
BSc Hons Computer Science GPA: 3.99 (Outstanding Graduate, Academic Excellence Scholarship)	

RESEARCH EXPERIENCE

Real-Time Perceptual Spatial Quantization for Foveated Rendering	2025.12 - 2026.04
<i>EGSR2026 (Accepted)</i> Visualization and Computer Graphics Lab , University of Leeds	
- Conducted a comprehensive literature review on color perception and gaze-contingent rendering. - Designed algorithm and workflow for gaze-contingent color bit-depth reduction for foveated streaming. - Conducted perceptual experiments to examine discrimination for spatial quantization. - Achieved 83% data bandwidth reduction compared to standard 24-bit true color frame.	
MRTodo: Synchronizing Paper and Digital To-Do Lists in Mixed Reality	2025.04 - 2025.09
VVAI Lab , Texas A&M	
- Designed and developed a context-aware productivity application. - Designed and executed user study, including both qualitative interviews and quantitative surveys to evaluate usability, effectiveness, and user satisfaction.	
Real-time Localization & 3D Scene Visualization	2024.07 - 2024.08
Advanced Display and Sensing Research Center , Yongjiang Laboratory	
Developed a web-based interactive tool to visualize real-time SLAM localization of camera pose and 3D scene reconstruction results.	
Transmogrifier Game: Visuomotor misalignment for Non-humanoid Avatars	2023.10 - 2024.05
<i>Undergraduate Dissertation</i> Mixed Reality Lab , University of Nottingham	
- Designed physics-based locomotion for non-humanoid avatars and movements of extra body parts through inverse kinematics and procedural animation. - Designed haptic feedback and visuomotor misalignment to simulate varied body weights and forces. - Simulated vision features including color perception, echolocation, blurring vision. - Conducted user study and semi-structured interviews, gathered qualitative data exploring body ownership.	
Fire Safety Training Through Virtual Reality Serious Gaming	2022.06 - 2022.10
China Beacon Institute, University of Nottingham	
- Designed and developed fire extinguishing equipment training in non-immersive and immersive settings. - Designed experiment workflow for analyzing the effectiveness of different simulation in terms of learning outcomes and user engagement and evaluating interaction preference of users.	

SKILLS

Languages: C#, C++ (OpenGL, Vulkan), Python, JavaScript
Tools: Unity, Unreal Engine, Figma, Adobe Photoshop, Blender